

About Vi-Fi (“Visible - Findable”) - how to describe and to search sites with standard and universal semantic-pragmatic tree: Semantic Web as Pragmatic Web

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ABSTRACT

The new approach to Web search, which provides powerful narrow-focused marketing tool for small on-line businesses as well as for everybody, who want to increase his visibility in Web, is proposed. The core of approach is observable (small enough - of the order of tens) standard universal system of both attributes and their values, which describes requests of WEB-users and content (offers) of sites in the same language and allow to calculate easily congruency between query and site. This system is based on pragmatics (logic of customer's request) rather than on usual (“pure”) ontology and is organized as tree. Both web-masters and searchers browse this tree to describe their sites and queries respectively. The tree changes until take its ultimate form as approach is realized.

Categories and Subject Descriptors

I.2.4 [Artificial Intelligence]: Knowledge Representation Formalisms and Methods – *representation languages*.

General Terms

Human Factors, Standardization, Languages.

Keywords

Web search, Semantic Web, Pragmatic Web, ontology, semantic tree, standard tree, universal tree.

1. INTRODUCTION

View from the searcher's side. There are things that may be easy found in Web, and the things which are very difficult to find. It is difficult even to understand either they present in Web or not. For example, right now it was very simple for me to find some information about Semantic Web. I made search in Google and found article in Wikipedia with a lot of references, some of them directed me in W3C site etc. But it was simply impossible (at least for me with my experience of Web search) to find somebody who proposed or developed the approach similar to my own. It was even impossible to find out that nobody elaborated these ideas. If I want to find notebook Toshiba Satellite 2060CDC, it would be very simple for me to resolve this problem. But if I want to find cheap, convenient, reliable notebook suitable for my personal aims, I will need to spend at least some days studying market, searching consumer's forums etc. Or alternatively, I must buy

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something based on advice of friends or simply something “famous”. The same problem I meet when I'm going to travel. This is very simple to find some hotel in every city. But it is very difficult to find “right” hotel for you. And of course, it is almost impossible to find more or less significant like-minded group, which shares your views, if these views are not very common. And so on. To say simply I can find in Web something, that “Big Boys” want to sell me, rather than what I need indeed.

View from the seller's side. If I propose some specific things for specific people, I have very few chances that I will find these people or they will find me. If I have specific area of expertise and want to sell (or even propose free) it to somebody, I have chance (at best) to find only very small friction of my potential clients. The same thing is with search friends, collaborators etc. But in all these cases my proposition will reach a lot of people who do not need it. In other words, if I am not “Big Name”, I am invisible on Web.

Of course, there are a lot of attempts to deal with these problems, like EBay Auctions or Yahoo Questions, but all of them cannot resolve the problem of satisfactory matching offers and requests on Web. For example, there are hundreds or thousands places more or less similar to what I need but there are no tools to choose from them one or two most appropriate.

2. APPROACH

2.1. General consideration

The roots of described problem are quite obvious: if you say very briefly of what you want, then you have not too big chance that your interlocutor to understand you, especially when you want something special. If you try to minimize your query or description of your site, you cannot find what you do need (but what you *did not* say) or you cannot be visible by those who do need what you offer (but again what you *did not* describe). Thus, to resolve this problem both searchers and web-masters must present their needs and offers respectively in all *necessary* details *explicitly*. But this is only one, necessary, but not sufficient condition. The second (“more sufficient”, but again not “completely sufficient”) is that they must use the *same* language.

Ultimate solutions of our problem might be imagine as model of Semantic Web [1] based on Natural Language Processing (NLP), where the user formulates his query in natural language, the semantic analyzer extracts the meaning of this query and after that semantic search engine analyses content of sites to find ones which are appropriate for query. However this idea may be realized today (and perhaps not just today, but in principle) only for very limited scopes of application. I will not discuss here all difficulties, which arise when we attempt to create NLP-based Semantic Web - they are well known (for example, [10], or [13]),

but I want to point at only one of them, which attracts less attention of developers.

Indeed, the meaning of request often cannot be extracted from text of query, in principle. If somebody asks for “hotel in Florida” he may simply not state explicitly that he need hotel for holiday with his family. Thus, special procedure is necessary in many cases to clarify request. At the same time, not every hotelier put in his web-site that his hotel especially suited for leisure. Sometimes he does not do this intentionally - being afraid to reduce the number of potential customers (for example, business travelers may want more specialized for their needs hotel, rather than his “leisure hotel”), but sometimes he simply do not think in logic of his potential customers. In the last case he may need special procedure, which helps him to show in his site all aspects of his hotel, which may be potentially crucial for his customers.

In their attempts to overcome problems of NLP-based approach the developers of Semantic Web choose another way (for example, [12]). They create the systems of interrelated concepts (“ontologies”) and implement semantic information into description of sites. This approach works, but it also related with many limitations and problems. First of all, ontologies, which really work today, are quite specific and cover only narrow “scope of World”. Second, the languages of semantic description of sites are rather complex, and to describe his site “the lay man” must invest a good deal of efforts to learn necessary language. And “the lay user” must invest the same efforts to use Semantic Web without intermediary “brokers”. Moreover, both must learn not one, but many different “ontologies” and corresponding languages. But even in the case, when the lay person decides to become expert in Semantic Web, he has no guarantee, that his efforts will result in some advantages, because he easily discover the absence of ontology, which reflect namely his own semantics – his meaning, his intentions and so on.

The central idea of my approach is rather close to general idea of Semantic Web. If both sites and queries are described in the same formal language, then we may make the search more focused and make the sites more visible. Two simple considerations are in a core of this approach. If I offer to customers exactly what thousand other seller offer, my chance to be “findable” is one thousandth in best. And if I formulate my need like millions others formulate their needs, I cannot expect, that I find the site which is right namely for me. But if I describe my need in some details and in the *same* language which sites’ owners use to describe their sites, my chances to find what I need increase as well as chances of site’s owner to be found by me increase also.

But every speaker of some natural language has his/her own experience and this results in that he/she has his/her own “model of world” and uses his/her own language with his/her own *meanings* of words. And although the words themselves are common for all speakers of this language, the different meanings of the same words for different speakers (and listeners) make their similar sounding words, in fact, different ones. To say the same thing in other words, each person has his/her own system of concepts, and in this sense, has his/her “own language”. Thus, the task is to replace (for aim of Web-search) all this different individual “languages” by one common language. One way to resolve this task is to introduce one standard ontology and to construct one artificial “language of concept” to force users to describe themselves in framework of this artificial ontology with help of its artificial language, rather than to allow them to use

their own (“natural”) models of world, systems of concepts and languages, with hope to extract the correct meaning from these natural statements.

However, although the overall approach is in frameworks of Semantic Web, its novelty is determined by the character of its ontology:

- **by the very limited quantity of objects (classes) and attributes;**
- **by the rather common (universal) character of attributes;**
- **by the pragmatic rather than “pure semantic” nature of attributes; and**
- **by the rejection from attempts to build “the only right ontology” in favor of self-developing ontology.**

What kind of artificial language I mean? The simplest and most natural (in logic of developer of such system) language of this kind is the language of attributes/values (systems of concepts). If we propose for simplicity, that all attributes has the same number of values, then the words in this language are the matrix $\{x(i,j)\}$, where $x(i,j)$ is meaning of predicate “*i*-th attribute of object has *j*-th value”. To make our system of concepts complex enough to produce tens and hundreds of billions different words, corresponding to different types of queries and sites, we must introduce some more or less *independent* attributes, which describe both queries and sites.

For example, if we introduce 10 independent attributes, each of them may have 1 of 10 different values, the total variety of different types of queries/sites will be 10^{10} . Thus, it is enough for searcher (web-master) to answer on 10 simple questions to make his query (site) different from 10 billions other queries (sites).

Of course, this situation is some model over-simplification. The real set of attributes and their values is organized in a more complex way - as hierarchy (tree), where the upper levels correspond to more general (abstract) concepts and lower ones correspond less general (abstract) concepts, which disclose meanings of upper concepts. In other words, each complex attribute itself has its own tree-like structure, which is similar to ones we use when we browse in a book shop searching the books of specific theme. But even taking into consideration over-simplification of example above, the estimation “Ten-ten” (ten attributes, each with ten values) gives right ideas of both complexity of descriptive language and necessary efforts to use it. Since there are no any problem to elaborate standard and simple procedures for describing sites and forming queries (for example, with using dialog wizards), the approach seems to be rather practical.

Thus, the present approach does not aim to elaborate detailed ontology of Universe with huge volume of objects, classes etc., but intend to introduce rather limited set of common (universal) categories, which must be sufficient to match demands and offers, but of course which is not enough to describe both sites and queries in **all** details.

And the last (by order but not by importance - indeed, it is VERY important) question is about the character of our attributes. To make approach practical our concepts must reflect searcher’s

pragmatics (in the meaning how this concept, “pragmatics” is determined today in philosophy and linguistics [9, 14]), i.e. what searcher needs and what he wants to get from site (to find in Web). In other words, both queries and sites must be described in the logic of searcher’s request. In particular, web-master describes what potential user may find valuable for him in the site.

To describe user’s pragmatics I use 3 groups of descriptors, which describe:

- (a) the needs (goals) of user,
- (b) the activity of user, and
- (c) the user himself in terms of his potentially relevant attributes and social roles.

Below I provide examples of both the tree of concepts and the simple language for describing sites and queries. But in the beginning I’d like to discuss more “philosophical” topic.

2.2. Semantic Web and Pragmatic Web

Historically the development of theoretical linguistics came from discoveries of lower levels of the organization of text to discoveries of more and more high levels –from “pure linguistic” phonology and phonetic, morphology, syntax to “more psycholinguistic” semantics and pragmatics [2]. Artificial Intelligence (AI) repeats this philosophy when it divides the overall processing natural language into different levels (subsystems), each of them processes separately its own level of organization of text [5, 10, 11, 13]. However, studies in cognitive psychology (for example, [4], or [6]) demonstrate that human way to process texts often is quite different from this formal scheme. In context of our topic the important difference between “AI paradigm” of text processing and “Human paradigm” is that in AI semantic level is lower than pragmatic level, while cognitive psychology shows that the pragmatic analysis is primary. In other words, human being often moves from realizing general intention of the author of text as well as realizing overall communicative situation and respectively overall context to realizing details of text’s meaning.

There is a deep sense in this way to process text. The texts differ one from other by potential variety of their meanings. There are texts, where this variety is relatively low, for example, “the apple is on the table”. But the texts of this (‘first’) kind do not constitute the universe of all texts or even majority of all texts. There are a lot of texts, where *potential* variety of meanings is practically infinite. For example, in simplest “piece of knowledge”, which is formulated by the very simple utterance “good house”, the meaning of “good” may be very, very different: beautiful, comfort, convenient, large, cheap, reliable and so on and so on. To determine what meaning of such “text of second kind” is real we must know many things about the author of utterance: who is the author, what does he do, what are his goals, what is his situation (which determines the context of phrase) and so on. All this questions set the scope of pragmatics. Thus, pragmatics becomes the powerful tool to

reduce the *potential* variety of meanings to one *factual* (*actual*) meaning.

To do this pragmatic establishes the hierarchy of knowledge, i.e. hierarchy of objects (things of world) and relationships between them, where the upper levels consist of “subjective” knowledge about the author of utterance – his personality, his activity and his goals, and the lower levels is the knowledge about the world as it is determined by the knowledge about the author.

Today AI-systems (like PUNDIT, described in [8]) analyze the semantic level (i.e. concepts and relationships between concepts) of the text and only after that may start pragmatic analysis. The idea of the present approach is to reverse this situation and to move from pragmatic level, i.e. from analysis of the intentions of speaker and of his personal meaning of words and phrases, to “common semantics”, i.e. to “universal” meanings of *some* phrases, which are common for all speakers. This is the reason why in described below tree the common semantics appears only on third level of hierarchy (the branch 1.1.2 “ABOUT”) while the upper levels consist of descriptions of needs, activities, and “personal” attributes of the user.

In other words, this approach considers semantics to be a part of pragmatics rather than vice versa, how it is commonly accepted today. And if we intend to create Semantic Web we must start to do Pragmatic Web.

But how must we do Pragmatic Web, when there is at least one (but very serious) reason why usually semantic analysis is made in early stage and serves as base for pragmatic analysis? Indeed, computer systems analyze text beginning from simple and small units and use results of this preliminary analysis to analyze bigger and more complex units, because this “from down to up” is the only way they can reach the level of complex units. If we do not work in level of concepts, meanings, relationships, how can we manage to work in level of context, intentions, goals etc.? And indeed this is impossible, if we try to build analyzer, which works only with text without interaction with human being. But everything is changed as we include in our system a man. We have not too big chances to make good pragmatic analyzer based on results of semantic analysis only. But when we include man in our system our task become much easier. Now we may receive more or less exact pragmatic data directly from their subject. This is why Pragmatic Web must base on man-machine interaction rather than on “pure computer” semantic/pragmatic analyzers.

3. STANDARD UNIVERSAL SEMANTIC-PRAGMATIC TREE

Standard universal semantic-pragmatic tree (SUSPT) consists of attributes and their values. Attributes may be simple or complex.

Complex attributes is subset of set of attributes, which may be also simple or complex (but, informally, even if “daughter” attribute is also complex, it is “simpler”, than “mother” attribute).

Simple attributes are determined by their values, which may be “simple values” (in present model version of SUSPT all simple values are Boolean, but they may be also numbers, strings etc., for example, attribute PERSON may have string-value name, or number-value age) or tree-values. Tree-value is described by own attributes, which again may be complex or simple, and the last again may have simple values or tree-values, and so on, until we reach simple attributes, which has only simple values.

In scheme below the simple attributes are written IN CAPITAL LETTERS, complex attributes are written in **BOLD CAPITAL LETTERS**, “simple values” are underlined and tree-values are written **IN CAPITAL LETTERS, BOLD AND UNDERLINED**.

In highest level of present version of SUSPT there are 6 attributes, which together describe searcher’s request. This 6 attributes are divided in 3 groups:

- What does searcher needs - 2 attributes;
- What does searcher does (to describe for what kinds of searcher’s activity he may need what site offers) - 3 attributes;
- Who is searcher - 1 attribute.

1. NEED (what does site (page) offers or what does searcher need)

{**INFORMATION**, expert, teacher, counselor, employee/worker, client/consumer, business partner, friend, relative, adoptive relative, organization, service/work, goods, files, money, job, other} (17 /16; 1-1; 1-5) (17 values; 16 of them are “simple values”; not less than 1 and not more than 1 (i.e. 1 exactly) must be chosen for query description; not less than 1 and not more than 5 must be chosen for each *unit* (see below) of site description. Here and below all constants, which determine how many variants must be chosen, are for the aim of illustration only.)

1.1. **INFORMATION** (tree-value)

{**TYPE**, **ABOUT**} (2/0; 2-2; 2-2)

1.1.1. **TYPE** (simple attribute)

{news, opinion/view/theory, reference, personal story, other} (5/5; 1-1, 1-4)

1.1.2. **ABOUT** (complex attribute)

{**LOCATION**, **TIME**, **IDEALITY**} (3/0; 1-3, 1-3)

Category ABOUT describes objects of world, information of which is offered by site (is searched by consumer (searcher)). To describe different objects 5-dimensional Model of World is proposed, in which to three dimensions of spatial world (LOCATION) two other dimensions are added. First of them is time. The second one has no conventional name. I named it “Ideality”. It is about ideal or “subtle” entities of World beginning from concepts (which determine classes of material things), relationships, ideas, mental states and up to super-natural creatures like angels, demons and after all - God.

1.1.2.1. **LOCATION**

{space, sky, bowels of the earth, ocean, geographical region, group of persons, artificial things, other} (8/8; 1-2, 1-2)

1.1.2.2. **TIME**

{past, present, future, eternal, other} (5/5; 1-2; 1-2)

1.1.2.3. **IDEALITY**

{class/concept, ideas, activity, arts, mental/spiritual states/processes, subtle worlds, super-natural beings, other} (7/7; 1-3; 1-3)

2. **PRICE**

{free, money - cheap, money - average, money - expensive, loan, rent, change, work, other} (9/9; 1-3; 1-3)

3. **OBJECT OF CONSUMER (SEARCHER)’S ACTIVITY**

{ **PERSONS**, space, sky, Earth as whole, oceans/seas, rivers, land, animals, technical system, art, science, programming, writing, other } (14/13; 1-2; 1-2)

3.1. **PERSONS** (tree-value)

{**GROUP**, **ASPECT**} (2/0; 2-2; 2-2)

3.1.1. **GROUP**

{myself, family, acquaintances (friends/colleagues/staff), party/union, society/people, confession, humankind, other} (8/8; 1-2; 1-2)

3.1.2. **ASPECT**

{mental-spiritual state, values, relationships, behavior, legal state, wealth, health, other} (8/8; 1-3; 1-3)

4. **TYPE OF CONSUMER (SEARCHER)’S ACTIVITY** (or searcher’s one)

{work/business, household, family, rest/holiday, health recovery, self-development/spiritual growth, other} (7/7; 1-3; 1-3)

5. **CHARACTER OF CONSUMER (SEARCHER)’S ACTIVITY** (or searcher’s one)

{routine, creativity, improving, learning/training, exploring/investigating, game/play, love/help/care, fighting/destroying, artistic reflection, other} (10/10; 1-3; 1-3)

6. **CONSUMER (SEARCHER)**

{**AGE**, **EDUCATION**, **FAMILY**, **HEALTH**, **LANGUAGE**, **OCCUPATION/PROFESSION**, **NATIONALITY/RESIDENCE**, **RELIGION**} (8/0; 0-8; 0-8)

6.1. **AGE**

{ ≤7, 7-12, 13-19, 20-28, 29-40, 40-55, 55-70, ≥70 } (8/8; 1-1; 1-6)

6.2. **EDUCATION**

{less than high, high, university, degree} (4/4; 1-1, 1-3)

6.3. **FAMILY**

{wife/husband, children, grandchildren, parents, grandparents, sisters/brothers, other} (7/7; 1-7; 1-7)

6.4. **HEALTH**

{chronic decease, serious illness, temporary illness, healthful,

other } (5/5; 1-2; 1-3)

6.5 LANGUAGE

{English, Spanish, Russian, Other European languages, Chinese, Indian languages, Arabic, other} (8/8; 1-6; 1-7)

6.6. OCCUPATION/PROFESSION

{computers, science, engineering, finance, law, medicine, sport, media, arts, politics, business, civil administration, military, other} (14/14; 1-3; 1-13)

6.7. NATIONALITY/RESIDENCE

{USA, Europe, Canada, Japan, Australia/New Zealand, Latin America, China, India, Africa, Muslim World, Russia/CIS, other} (11/11; 1-3; 1-10)

6.8. RELIGION

{Western Christianity, Eastern Christianity, Islam, Buddhism, Hindu, Jewish, other} (7/7; 1-1; 1-6)

4. DESCRIBING QUERIES AND SITES

4.1. Describing queries

Each query is described by its word-tree, which determines the subset of set of all leaves of the SUSPT, which correspond to this query. Word-tree may be determined recurrently as string $T = (T^1 \dots T^N)$, where T^i ($i = 1, \dots, N$) is Boolean constants 1 or 0, or word-tree.

Thus, word-tree is word in language of description of trees (LDT) with alphabet from 4 terminal symbols { **1** ; **0** ; (;) } and one non-terminal symbol S and grammar: $S \Rightarrow (S)$; $S \Rightarrow S(S)$; $S \Rightarrow (S)S$; $S \Rightarrow 0S$; $S \Rightarrow 1S$; $S \Rightarrow S1$; $S \Rightarrow S0$; $S \Rightarrow 0$; $S \Rightarrow 1$

Each SUSPT determine structure of all words of its LDT. For example, version of SUSPT, described above, determine the following structure:

((((X1.1.1[5]) ((X1.1.2.1[8]) (X1.1.2.2[5]) (X1.1.2.3[7]))) X1[16]) (X2[9]) (((X3.1.1[8]) (X3.1.2[8])) X3[13]) (X4[7]) (X5[10]) ((X6.1[8]) (X6.2[4]) (X6.3[7]) (X6.4[5]) (X6.5[8]) (X6.6[14]) (X6.7[11]) (X6.8[7])))

(To make structure easier readable I used brackets of different size for different levels of tree.) Where $Xn^1 \dots n^k [j]$ - is (Boolean) vector (of size j) of simple values of simple attribute, corresponding to vertex $n^1 \dots n^k$ of k-th level of SUSPT.

Word-tree may be recorded in full (or "normal") form as it is shown above or in brief form. In brief form the full record of structure of attributes or tree-values, which are not used for description of particular query (site), are replaced by Boolean "0" (zero). More specifically, the meanings of elements of brief record of word-trees which describe queries are:

"0" in position of attribute means, that this attribute is not appropriate (not important) for query;

--"0" in position of value of attribute means that this value is not appropriate for query;

---- Word-tree in position of tree-value of attribute is tree-value of

this attribute;

---- "1" in position of simple value of attribute means that this attribute has this value.

Example of brief record

(000(100000)(0000000001)((00001000)(0010)000000)) - is the brief form of record of word-tree for following query:

type of request, price and object of searcher's activity are not important, type of searcher's activity is "work or business", character of searcher's activity is one, which are not reflected in SUSPT (may be determined additionally for development of SUSPT), age of searcher is "29-40 years", education level is "university degree" (bachelor or similar), other personal characteristics of searcher are not determined.

4.2. Description of sites (pages)

Description of sites (pages) consist of a number of units, each of them corresponds to some sets of queries, which site (page) address to. Site-wizard helps web-master to form as many units as he consider to be sufficient to describe his site (page).

Each unit of description of site (page) is similar to description of query with one essential exemption in its semantics. Zero ("0") in position of attribute in brief record of query means that this attribute is not appropriate for this query. In the description of site zero ("0") in position of attribute means that the site has offers for all values of this attribute. For example, if webmaster ignores attribute "age of customer" he means that this site addresses to customers of all ages (from "younger than 7" till "older than 70").

Sites differ one from other by their universality, i.e. by how reach their content and to how many users (different types of queries) site addresses. To stimulate web-masters describe their sites to be addressed to reasonably narrow range of customers, the search procedure must give first places to more specific, narrow-focused, specialized sites.

4.3. Language and family of languages

Among standard Web languages XML seems to be appropriate for describing both queries and sites. The simplest option to describe site is the following single XML statement:

<LDT v.MM.DD.YY> [tree] | <!--[other-1]--> | ... | <!--[other-K]--> </LDT>

Where [tree] is word in LDT and [other-i] is text string, which user enters together with i-th "other" (from left to right into the code of tree).

5. SEARCH AND METRICS

Result of search is determined by comparing description of query with description of site (page). Congruence between description of site (page) and description of query is determined by metrics on $\{T\} \times \{T\}$. This metrics may be introduced by many ways. Below (mainly as example) I provide one class of such metrics, which seems to be practical. Calculating measure of congruence in this metrics consist of 3 steps.

I. Transforming records from brief form to "normal" form

For this:

1. In descriptions of sites

a) each “0” in position of attribute must be replaced by full record of the word-subtree of this attribute, consisted only of “1”s. For example, in description of site (page) “0” in position of attribute 1.1.2. (**ABOUT**) must be replaced by ((11111111)(11111)(1111111)) [strings of “1”s for 3 sub-attributes of **ABOUT** - LOCATION, TIME and IDEALITY].

b) each “0” in position of tree-value must be replaced by full record of the word-subtree, consisted only of “0”s. For example, in description of site (page) “0” in position of tree-value 3.1 **PERSONS** must be replaced by ((00000000)(00000000)) [strings of “0”s for 2 simple attributes of **PERSONS** - GROUP and ASPECT].

2. In description of query:

a) each “0”s in position of attribute must be replaced by full record of the word-subtree for this attribute, consisted only of letters “N”. For example, in description of query “0” in position of attribute 1.1.2. (**ABOUT**) must be replaced by ((NNNNNNNN)(NNNN)(NNNNNN)) [strings of “N”s for 3 sub-attributes of **ABOUT** - LOCATION, TIME and IDEALITY].

b) each “0” in position of tree-value must be replaced by full record of the word-subtree, consisted only of letters “N”. For example, in description of query “0” in position of tree-value 3.1 **PERSONS** must be replaced by ((NNNNNNNN)(NNNNNNNN)) [strings of “N”s for 2 attributes of **PERSONS** - GROUP and ASPECT].

Thus, in result of transforming brief form of record into normal form descriptions of query may become 3-valued instead 2-valued (Boolean) string.

II. Determining similarity between query and each unit of the description of site (page)

In given model metrics I introduce only 2 values for measure of similarity between query and site (page) - “similar” or “different”. In principle, measures with many values may be easily introduced.

1. Simple attributes of query and of one unit of the description of site are similar, if and only if they have:

- a) not less than n common “1”s (not counting value “other”, if attribute has value “other”); and
- b) not less than m common “0”s.

2. Complex attributes (tree-values) of k-th level of query and complex attributes (tree-values) of k-th level of unit of the description of site are similar if and only if they have:

- a) not less than p^k similar attributes and/or tree-values of (k+1)-th level, and
- b) not less than n^k common “1” (not count “1” for value “other”, if attribute has value “other”), and
- c) not less than m^k common “0”.

The meaning of demand for some amount of common “0”s in descriptions of query and site is to prevent from too common descriptions of sites, which may be made with aim to attract as many customers as possible.

III. Determining measure of congruence of query and site (page)

Site is congruent to query if at least one of units of description of site is similar to description of query.

In course of development of the approach it will be possible not only use different metrics, but even provide searcher option to determine his own metrics from some class. It will increase flexibility of overall approach essentially and in some cases will help user to better understand what he need.

6. IMPLEMENTATION AND PROBLEMS

Described approach aims to be used as addition to existing search portals rather than as alternative to them - as one more option of “Advanced search”. I do not see any technical problems to realize it: wizards, parsers and supplements to search programs all are very simple from programming point of view. But nevertheless there are two serious problems with this approach.

The first problem is to find the right system of attributes-values which provide more or less even distribution of real sites (queries) among all possible billions variants of descriptions. To resolve this problem SUSPT must not be static (“dead”, once and forever established), but must be dynamic, **alive** tree - tree, which grows, blossoms and fruits. The experience of work with the hierarchy of concepts must result in further evolution of hierarchy, which may include:

- elimination of one attributes and/or values or even whole subtrees, which show themselves as empty (no sites registered in the category) or rare used by searchers;
- dividing categories, which describe many sites or are often used by searchers, into subcategories;
- introducing new categories, based on propositions of users, which they enter together with value “other”.

The necessary statistic programs for this analysis also seem to be rather simple. Of course, web-masters will be provided with tools for updating their sites in accordance with new versions of SUSPT.

The second problem is that people do not only use different words to say the same thing, but also use different generalization for the same things. For example, the trip to resort with family for one man will be part of his activity “recovery” or “care for health”, but for other - part of his activity “love to children”. To say more formal, each value of some attributes determines own set of things of the World, and there are significant intersections between such sets. Less formal - there are a lot of things which are described by different values of the same attribute. The problem is to eliminate effect of these intersections.

The difficulties of attempts to resolve this problem are approximately the same as the difficulties with other “semantic tasks”. For example, we may try to introduce such values, which have minimal intersections. But in this case often we may be forced to introduce too many different values.

More or less practical method to deal with this problem in spirit of present approach is to predetermine relationships between phenomena and categories with aim to limit as much as possible users' freedom to determine these relationships themselves. To do this we may provide for every category (attributes or tree-value) a list of examples - what given category includes, for example, together with displaying (highlighting) category "health recovery" user get the list of actions, which include in this category: "rest", "psychological relaxation", "search for physician", "search for diet", "exercises", "eliminating factors of risk", "search for pharmacy", and so on and so on. Of course, such measures cannot resolve this problem completely - unfortunately, there are TOO many phenomena in world! But we may hope to "grow" our artificial language, including into the language new and new concepts and, at the same time, keeping its "artificiality" and, in particular, trying to minimize intersection between concepts (i.e. to keep tree-like structure of growing SUSPT and not allow its branches to interweave).

7. CONCLUSION

It works, or it does not work? "That is the question" [3]. Since I described the approach, which is not realized yet, the answer is definitely "no" - this approach does not work. It does not work **yet**. And of course, as any approach, which is not realized yet, it is possible to anticipate, that it will not be realized at all, or it will be realized so poorly, that its realization will not work. But I'll try argue, that this approach may be realized in such a way, that it will do work.

First of all, the systems of categorization similar to SUSPT (although not so "universal") exist and work. For example, the format of present article includes *Categories and Subject Descriptors*, which is (are) one or more leaves of some "concept tree", which aims to describe all possible topics of scientific papers. This tree together with similar trees for different subjects may be considered as branches of SUSPT.

"Works or does not work" depends on many factors. Besides "subjective", or "organizational" factors (for example, chances to implant such project in existing large-scale search business), first from these factors is the possibility to create SUSPT which generate billions classes of web-pages and each of these classes includes not more than, say, 10 individual pages. I do not expect that one developer may create such SUSPT himself. But I do believe, that such SUSPT may be evolved in process of interaction between users and developers of overall system, as I described in section 6. Of course, this does not guarantee that website, which is correspond to user's query will respond completely to user's need, but it makes less probable that necessary site, if it exists in web and described in LDT, will not be found.

The second factor is the correspondence between queries and sites which includes in the same class (leaf of SUSPT). For example, it is easy to propose that the person who does need *information* will categorize his need as search of *expert* or *counselor*. But this is very common problem for all languages (even not only for artificial ones): to describe what you want by means of some language you must be master of this language. (I do feel this too well at the present moment trying to express my thought in foreign for me English.) Thus, to eliminate this factor the users must learn LDT, and respective programs must include the means, which help users to do this. I said already that one of such means

may be prompts which explain the meaning of some category, for instance, examples of objects of this category and their difference from objects of other categories. For example, such prompt may explain, that *information* means information of common interest, which may be found in encyclopedias, text books and so on, that *expert* provides impersonal information, but besides helps to determine what information you need, while *counselor* provides personal advices.

I elaborated present approach with aim to introduce it in the major search engines like Google or Yahoo (or in similar national specific universal search engine) as one more option of "Advance Search". However, the same approach may be easy modified for more specific searches, for example, for book search in online shops like Amazon. Of course, in this case "Universal tree" must be transformed in "Specific tree". For example, in case of book shop, it will be tree of users' needs and expectations. Here we have some point for comparing, since we know that the only searches of books which works satisfactorily today is search of titles, authors, keywords, ISBN, but not search of book's ideas, or book's significance for reader. Thus, every attempt to introduce semantic-pragmatic search in this field do not meet competitors today. By the way, the similar situation we have and with universal search engines.

And at the end I would like to emphasize two strong features of the present approach, which also strengthen the hope that this approach may work. The first such strong side is the high motivation of both owners of websites and their potential users to spend their time and efforts to make adequate descriptions of websites and adequate queries respectively. Today there are a lot of systems for "Semantic search" [7], but of course all of them work with their own sites (pages), with their own specific ontology and with descriptions in terms of this ontology which are made by developers of particular system of "Semantic Web". The present approach proposes to shift responsibility for semantic descriptions of the sites on sites' owners and by means of such "collaboration" to enlarge the group of developers crucially. In result we get possibility to put the idea of Semantic Web on absolutely different scale.

The other strong side of the present approach is principal enlargement of the function of Web itself. Today it is practically impossible to find in Web the answers for question which is not very common, but specific for given person. The reason for this is practical *impossibility to find* the right information (or right expert) rather than *the absence* of necessary information (expert) in Web. In this situation users even do not attempt to find the answers on questions which are often very important for them. Implementation of the present approach opens new opportunities for them. On the other hand, this system will stimulate providers of Web services (for example, experts of all kinds) to propose not what hundreds other providers propose, but something specific what is in real demand of users.

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